Immanuel Church, Seatides

PROJECT NO 9721

Project Specification

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0.0 GENERAL

0.1 This project specification, together with the relevant structural Engineering drawings, should be issued to all relevant sub-contractors by the main contractor.

0.1 All **dimensions and levels** to be checked on site and correlated with Architect's drawings and details before construction commences and any discrepancies are to be reported immediately to the Architect and Engineer.

0.2 All **waterproofing details** are to be in accordance with the Architect's (or his/her appointed specialist) specifications and instructions. The structural Engineer is not responsible for any waterproofing or waterproofing details whatsoever.

0.3 All details and dimensions shown on structural drawings are subject to confirmation on site and during construction.

0.4 All construction methods and materials used are to be in accordance with the requirements of **SANS 10400** and **SANS 01200** and all other applicable SABS codes of practice available from the SABS at contact no 012 428 7911. A non-inclusive list follows:

(i)	SANS 1200 A (1986):	General
(ii)	SANS 1200 AA (1986):	Small works
(iii)	SANS 1200 D (1986):	Earthworks
(iv)	SANS 1200 G (1982):	Concrete (structural)
(v)	SANS 1200 GB (1984):	Concrete (ordinary buildings)
(vi)	SANS 1200 H (1990):	Structural steelwork

Should the contractor's **appointment be based on the newer SANS 2001**, the SANS 1200 references should be replaced with SANS 2001:

(i)	SANS 2001-BE1:2008	Earthworks (general)
(ii)	SANS 2001-BS1:2008	Site Clearance
(iii)	SANS 2001-CC1:2012	Concrete works (structural)
(iv)	SANS 2001-CC2:2007	Concrete works (minor works)
(v)	SANS 2001-CM1:2012	Masonry walling
(vi)	SANS 2001-CM2:2011	Strip footings, pad footings & slab-on-the-ground foundations for masonry walling
(vii)	SANS 2001-CS1:2012	Structural steelwork
(viii)	SANS 2001-CT1:2011	Structural timberwork (flooring)
(ix)	SANS 2001-CT2:2011	Structural timberwork (roofing)
(x)	SANS 2001-DP1:2011	Earthworks for buried pipelines and prefabricated culverts
(xi)	SANS 2001-DP2:2010	Medium pressure pipelines
(xii)	SANS 2001-DP4:2010	Sewers
(xiii)	SANS 2001-DP5:2010	Stormwater drainage
(xiv)	SANS 2001-DP6:2012	Below-ground water installations
(xv)	SANS 2001-DP8:2011	Pipe jacking
(xvi)	SANS 2001-EM1:2007	Cement plaster

0.5 It is the contractor's responsibility to ensure that he/she and/or his/her **appointed sub-contractors** understand and comply with all relevant Engineering drawings and specifications and that he/she and/or his/her sub-contractors are adequately experienced to undertake all aspects of the work safely and correctly according to the relevant codes of practice.

0.6 All details and dimensions pertaining to any **existing structures** are to be confirmed on site by the contractor and the Engineer is to be immediately informed of any unexpected aspects pertaining to them. Where trying into existing structures or services, the contractor should ensure that the level **datums** are exactly the same.

0.7 A **benchmark, gridlines and site boundaries** are to be set out on site by a qualified and professionally registered land surveyor who has adequate professional indemnity cover.

0.8 All **proprietary products** specified for use are to be used strictly according to manufacturer's instructions and specifications at all times. If an alternative product is proposed, the contractor shall submit the proposal to the Engineer for approval. Relevant product specifications and/or data sheets may be called for if required by the Engineer.

0.9 The contractor is at all times to be fully responsible for **quality control** on site ensuring strict compliance with all drawings, details and specifications issued for construction by the professional team. This quality compliance extends to the contractor's sub-contractors. The onus is on the contractor to carry our representative sampling to ensure compliance to specifications. In the absence of a specification/s, the contractor should request such information from the Engineer.

0.10 The contractor is to comply at all times with all relevant **municipal regulations and bylaws** in the area of the site and is to ensure that he/she has a set of approved building plans on site at all times.

0.11 Should the Engineer be required on site to undertake **inspections or ad hoc meeting**, preferably at least 24 hours' notice is requested.

0.12 Sealing compounds for the building & construction industry to be in accordance with the requirements of:

(i) SANS 11077 (1984) sealing compounds for the building and construction industry, two-component, polyurethane-base

0.13 Sika-Pro 2HP universal one component polyurethane joint sealant (or similar approved) may be used on all wall joints and must be applied strictly to the manufacturer's specifications and the correct backing cord should be used where necessary.

0.14 Foundations and brickwork construction of single storey buildings, such as gatehouses to be in accordance with SAICE/istruct.e code of practice for foundations and super structures of single storey residential buildings of masonry construction as per NHBRC recommendations.

1.0 EXCAVATIONS AND FOUNDATIONS

1.1 The contractor is to refer to the site specific **Geotechnical report** to understand the soil conditions and water table of the site and its surrounds. During construction, the contractor is responsible for ensuring that all excavations are kept free of water at all times. The onus is on the contractor to plan and provide the necessary **temporary dewatering**.

1.2 Should the contractor request the use of the **permanent dewatering** drainage system during construction, the system is to be protected against all contamination and prior to handover the system should be comprehensively flushed.

1.3 The contractor is to identify and expose all **existing underground services** on site and prevent any damage to these services.

1.4 To ensure that the foundation level achieves the desired **bearing pressures**, all excavations (where not piled) are to be checked and approved by the Engineer before any concrete is cast. This approval will be delegated from the structural Engineer to the specialist geotechnical Engineer in many cases. The contractor should notify the Engineer or delegated geotechnical Engineer at east 48hours prior to requiring an inspection. Foundation excavations to be below any signs of organic material, roots, rubble, fill etc.

1.5 All final foundation sizes, depths and details are **subject to confirmation** on site by Engineer (see 1.4 above).

1.6 Assumed **safe bearing capacity** of soil under foundations – TBC. (Refer to Geotechnical report where applicable).

1.7 As soon as possible after the excavation has taken place (and approval of the foundation material has been given by the geotechnical Engineer) a 50mm **blinding layer** consisting of 10MPa mass concrete should be placed at the approved founding level. Excavations should not be left open for longer than absolutely necessary.

1.8 All foundations to be built on **firm in-situ material** (not on fill). In the case of any over-excavation, the difference in founding level should be made up with 10 MPa concrete unless otherwise instructed by the Engineer

1.9 The sides of all excavations and bulk cuts are to have **safe batters** as recommended in the Geotechnical report. In the absence of any data, a slope of 1v:1.5H should be assumed. In the event of excavations needing to be cut steeper than recommended or in the event of water ingress into the excavations causing collapse of the sides, **shoring** is to be provided (see also not 1.1 above). The contractor shall take whatever measure he deems necessary to protect all finished work once installed and any damage caused by the contractor or his subcontractor shall be made good at his own expense.

1.10 The contractor is responsible for providing the **design and construction** of the necessary shoring to all temporary excavations referred to above.

1.11 The contractor shall erect the necessary **hoarding** to ensure all deep excavations are safe at all times. The site is to be secured and access controlled to prevent the entry of unauthorised persons during and after working hours.

1.12 **Backfill** is to be undertaken in layers ensuring that replacement material matches the adjoining in-situ or Engineered layers (whichever applicable) to prevent differential settlement. Testing of the backfill material will be necessary (described elsewhere).

1.13 The top **level of all foundations** to be at least 300mm below final outside ground level or finished floor level, whichever is the lower, unless specifically noted otherwise on the structural drawings.

1.14 Foundations and brickwork construction of **single storey buildings** to be in accordance with SAICE/istructe code of practice for foundations and super structures of single storey residential buildings of masonry construction as per NHBRC recommendations.

1.15 All rising **foundation brickwork** to be constructed using an NFX brick and have continuous brickforce every 2nd course and cavity to be filled to underside of stepped DPC with 13mm aggregate, 15MPa concrete, taking care not to allow any honeycombing. Refer to the section on Brickwork in these specifications for further information pertaining to brickwork.

2.0 FORMWORK/TEMPORARY SUPPORT WORK

2.1 Forms shall have **sufficient strength** to withstand the pressure resulting from placement and compaction of the concrete and shall have sufficient rigidity to maintain the specified tolerances, the required shapes, finishes, positions, levels and dimensions shown in the drawings. The surfaces of forms which are to be in contact with the concrete shall be clean, free from deposits or adhering matter, ridges or spatter that will cause irregularities and blemishes to the concrete surface and shall also be free from indentations and warps.

2.2 **Surfaces** that are to be in contact with fresh (wet) concrete shall be clean and covered with a suitable coating material that will effectively prevent absorption of moisture, will prevent bond with the concrete and will not stain the concrete surfaces.

2.3 **Release agents** shall be applied strictly in accordance with the manufacturer's instructions and every precaution shall be taken to avoid the contamination of the reinforcement, prestressing tendons and anchorages. In the selection of release agents, due regard shall be given to the necessity for maintaining a uniform colour and appearance throughout on the exposed concrete surfaces.

2.4 It is the **contractor's responsibility** to ensure the adequacy of temporary support work (formwork) at all times. This responsibility to include, lateral bracing, warning signs and public protection measures. The contractor should ensure that all formwork is signed off by the formwork supplier and his own competent person prior to casting concrete.

2.5 **Soleplates** shall be provided when staging off material which is not suitable to support the applied loads. The contractor should ensure that the spacing of formwork legs and the size of the soleplates is such that settlement does not occur when loaded. The contractor (or his appointed sub-contractor) should obtain allowable bearing pressures from the geotechnical Engineer in order to carry out the soleplate design.

2.6 Support work to **remain in place** until the new structures is capable of providing its own support. Careful consideration is to be given to material curing times, sequencing, construction loads etc. When in doubt consult with the structural Engineer. See also notes on back propping.

3.0 CONCRETE

3.1 Concrete **characteristic 28-day strength** (if not stated otherwise on the drawings):

Mass concrete Blinding	15 MPa
RC Pile-caps (see note 3.4)	30 MPa
RC Bases (see note 3.4)	30 MPa
RC slabs, RC beams, RC stairs & RC landings	25 MPa
PT slabs, PT beams	30 MPa
RC Transfer beams	30 MPa
Surface bed	30 MPa
RC Cavity infill	15 MPa (using 13mm stone)
RC Retaining walls	25 MPa
RC Walls	30 MPa
RC Columns	40 MPa

3.2 All **concrete mix designs** are subject to approval by the structural Engineer. Cement to comply with SANS 50197. The choice of cements and extenders should be carefully considered for any given application. Weather conditions, timing of saw-cuts, post-tensioning stressing times etc are a few such examples.

3.3 Nominal **aggregate** size – 19mm, unless otherwise stated or approved.

3.4 All concrete structures cast below the water table to have a water cement ratio in the region of 0.5. The final mix design shall be submitted to the Engineer for approval but an allowable 60% (total) slag content may be assumed for tender.

3.5 **Concrete cube tests** to be carried out on all pours as per relevant SANS 01200 or SANS 2001 code (whichever applicable) requirements with specific reference to sampling frequency. The onus is on the contractor to closely monitor these results and report any strength shortfalls to the Engineer immediately.

3.6 All **precast concrete** suspended slabs to be in accordance with the requirements of SANS 1879 (2004): "precast concrete suspended slabs".

3.7 All performance tests for precast concrete floors should be in accordance with the requirements of SANS 09882 (1993): "Performance standards in building – performance test for precast concrete floors – behaviour under non-concentrated load"

3.8 **Curing and protection**: After formwork has been removed and as soon as it is practicable, all concrete shall, subject to the provisions of the relevant SANS 1200 or SANS 2001 be protected from contamination and loss of moisture by one or more of the following methods:

- Columns: **Wrap** in plastic sheeting immediately upon removal of formwork. The plastic wrap should be airtight. An approved **curing compound** is also acceptable.
- Beams: Spray with **curing compound** immediately upon removal of side shuttering, alternatively leave **shuttering** in place for 7 days.
- Slabs: Immediately after casting the concrete, it shall be protected from loss of moisture by one or more of the following methods:

  a) Covering the surface with **plastic sheeting** (preferable for hot and windy conditions)
  b) **Ponding** the exposed surfaces with water.
  c) Continuously **spraying** the exposed surfaces with water
  d) The use of an approved **curing compound** and in accordance with the manufacturer's instruction.

- Walls: Spray with **curing compound** or, alternatively, leave the **shutters** in place for 7 days.

Note: When using a **curing compound** for curing concrete the final surface finish is to be taken into account when choosing the specific product (consult with Architect). Certain curing compounds affect the adhesion of certain products (like painted parking lines on a wax based curing compound for example).

3.9 **Recommended earliest stripping time of formwork** is dependent on ambient temperatures, cement type and curing requirements. Actual times are to be discussed and agreed with the Engineer. An initial guideline for planning purposes is as follows:

Type of structural member or formwork	Duration or min concrete strength requirement
Beam sides and column sides	3 days
Wall shutters	3-7 days
Slabs soffits with props left underneath	4-7 days
Beam soffits and rib formers of a ribbed floor construction (both with props left underneath) **Stripping of props (single level suspended slabs where there are no back-propping requirements)**	7-14 days
Post tensioned flat slabs	minimum 30 MPa
Slab props including beam props	minimum 25MPa
Cantilevers	minimum 25MPa (but minimum of 14days)

For **multi-storey buildings back propping** is to be agreed with Engineer in detail and no stripping is allowed before permission or a specific method statement is obtained from the Engineer.

The **extent** of back-propping required is again dependent on the curing achieved on the support slabs below. For initial guideline purposes allow the following for suspended slabs:

- Slab being cast is fully shuttered propped
- 1st supporting slab to be 75% back-propped
- 2nd supporting slab to be 50% back-propped
- 3rd supporting slab to be 25% back-propped

Back-propping of transfer beams should be discussed in detail with the Engineer. In the absence of any detailed information, allow for the back-propping to extend down to the ground level.

3.10 No **core-drilling**, breaking of concrete, recesses or chases, other than those shown on the drawings or approved by the Engineer, may be cut or otherwise formed in the concrete without the permission of the Engineer.

3.11 **Fixtures**, hangers etc requiring fixing into or onto reinforced concrete is subject to approval by the Engineer. This includes all mechanical, electrical, wet services, data and other services fixtures to slab soffits and columns.

3.12 Subject to Architectural approval, all exposed corners for columns and walls in the basements to receive 30mm **chamfers** and all columns in brickwork walls to have no chamfers.

3.13 Concrete/cement seepage through shutters (burrs), spillage, overcast, highpoints etc are to be rubbed down and/or repaired to have a similar colour variation and texture finish to the adjacent concrete elements.

3.14 Permissible concrete **deviations and tolerances** are set out in the relevant SANS 01200 or SANS 2001 code (whichever applicable). The specific class and degree of accuracy shall be as per the bill of quantities and/or as per the specific technical documentation issued. When in doubt or if any discrepancies are found, the contractor should obtain an instruction on before executing the works.

3.15 The positioning of **cast joints**, where required in suspended slabs and beams, is to be discussed in detail with the Engineer. Typically cast joints should be cast at an angle of 30degrees to the vertical such that the portion of slab or beam closest to the column will support the adjoining slab/beam. This angled cast joint shutter should be formed using a solid shutter (not chicken mesh) to ensure that accelerated water loss (rapid curing and hence weakening) does not occur at this edge. Care should be taken to ensure that a neat straight cast line is achieved. The shutter may be removed no sooner than 3 days after pouring unless the adjoining slab is intended to be cast within 4 hours of said removal. The surface of the concrete shall be roughened (removing all laitance) using a wire brush.

3.16 The contractor shall take all reasonable steps to ensure that concrete is not poured during periods of **heavy or prolonged rainfall** without taking the necessary steps to protect such concrete against surface damage or water ingress. Additional water entering the mix can significantly weaken the concrete strength and affect durability and surface abrasion resistance.

4.0 REINFORCEMENT

4.1 Reinforcement shall **comply with** the relevant requirements of SANS 282, SANS 920, SANS 10144 and SANS 1024. See also the relevant section of SANS 10100-1 and SANS 10100-2

4.2 All placed reinforcement to be **checked and approved** by the Engineer before casting concrete.

4.3 The grade of accuracy for the cover over reinforcement shall comply with the requirements of the relevant SANS 2001-CC1 or SANS 1200. Reinforcement shall not be subjected to mechanical damage, rough handling, dropping from a height or shock loading.

4.3 The Engineer is to be given a minimum of 24 hours **notice** before reinforcement inspections and the work needs to 100% complete when he/she arrives for the inspection.

4.5 Minimum **concrete cover** to reinforcement (unless specified differently in bending schedules or drawings):

Foundations (pile-caps)	75mm (see bending schedule)
Foundations (bases)	50mm (see bending schedule)
Beams (internal)	30mm
Beams (external)	40mm
Columns	40mm (u.n.o)
Slab soffits and tops (internal)	30mm
Slab soffits (external)	40mm
Slab edge	40mm

4.6 The contractor must take particular care to ensure that the specified concrete cover to all reinforcement has been attained throughout and all rebar is securely and neatly fixed before the Engineer is called to undertake rebar inspections.

4.7 Bending schedules and fixing details for reinforcement will be issued at a later stage. **For estimation and planning purposes, the following high level reinforcing allowances may be assumed:**

Pile-caps	120kg/m³
Bases	80kg/m³
Retaining wall footings	100kg/m³
Retaining walls	110kg/m³
Shear walls/lift walls	160kg/m³
Columns	180kg/m³
Beams	130kg/m³
Transfer beams	175kg/m³
Slabs conventional	120kg/m³
Slabs Hybrid	100kg/m³
Rafts	115kg/m³
Cavity filled retaining walls	150kg/m³ (infill concrete)

4.8 All concrete is to be mechanically **vibrated and compacted**.

4.9 It is the contractor's responsibility to ensure that all reinforcement and cover blocks are correctly and accurately fixed, and **remain in place** during pouring.

4.10 The Engineer will **not** be responsible for any concrete cast where he/she was not notified to carry out a reinforcement inspection visit.

5.0 BRICKWORK AND MORTAR

5.1 All brickwork and plaster is to be constructed in accordance with the requirements of:

(i)	SANS 2001-CM1 - 2012	Construction works Part CM1: Masonry walling
(i)	SANS 0164 Part 1 & 2 - 1980	The structural use of masonry – Unreinforced masonry walling

(ii)	SANS 10249 - 2012	Masonry walling
(iii)	SANS 28 - 2010	Metal ties for cavity walls
(iv)	SANS 10164 - 2012	Masonry walling
(v)	NHBRC - 2015	Home Building Manual
vi)	SANS 10400	
vii)	The Concrete Institute	Relevant applicable leaflets

5.2 **Masonry units** shall satisfy the minimum specified compressive strength requirements and **comply with** the relevant requirements of one of the following standards

- SANS 227 for burnt clay masonry units,
- SANS 1215 for concrete masonry units.

5.3 Masonry Units – general **strength guideline**:

- Single storey construction (internal walling) - 3.5MPa
- External brickwork and double storey construction - 7MPa
- Structural and foundation brickwork - 14MPa (NFX in foundation if clay)

5.4 Masonry units shall

- have an average and minimum individual compressive strength as specified when five units are tested in accordance with the compressive strength requirements of SANS 1215;
- have dimensions such that the units can be built into walls within the joint tolerances, bond patterns and corner construction as set out in 4.4 of SANS 2001-CM1 - 2012
- not exhibit any surface pop-outs, should units contain slag, clinker or burnt clay aggregate;
- be of such a quality that, when delivered to the point of use, they are intact and have no corner chips of horizontal and vertical dimensions that exceed 15 mm;
- have, in the case of hollow units, face shells and webs not less than 25 mm thick, or one-sixth of the width of the unit, whichever is the greater;
- in the case of calcium silicate units, either have a demonstrated drying shrinkage of not more than 0,045 % or not be built into walls within 10 d of the date of manufacture;
- in the case of concrete masonry units, either have a demonstrated drying shrinkage of not more than 0,06 % or not be built into walls within 21 d of the date of manufacture; and
- have, in the case of burnt clay units, an irreversible moisture expansion of not more than 0,20 % and, in face masonry applications, be durable in the environment in which they will be constructed as evidenced by past performance over not less than 5 years or as reasonably demonstrated by other means.
- Masonry cements shall comply with the requirements of SANS 50413-1.

5.5 **Wetting** of masonry units

- Burnt clay units that have an initial rate of absorption that exceeds 1.8 g/m2/min shall be wetted 24 h before laying. Units shall be surface dry when laid. Immersion of the units in water shall not be permitted. (See SANS 2001 for testing guidance)
- Concrete and calcium silicate units shall not be wetted before laying and shall be laid dry. Stacks shall be protected from the rain.

5.6 The rate of new construction shall be limited so as to eliminate any possibility of joint deformation, slumping or instability, which might reduce the bond strength. As a guideline, the **height of walling** built in a day should generally not exceed 1.3 m to 1.5 m.

5.7 Unless otherwise specified, assume a class i **mortar strength** for all structural brickwork and a class ii mortar for non-structural brickwork. Mortar shall comply with SANS 2001 – CM1: 2012. No plasticisers are allowed without the approval of the Engineer. Where permission has been granted, plasticizers shall comply with the requirements of EN 934-3.

5.8 Mortar strengths:

- Class i = 10MPa (14.5 MPa preliminary laboratory test)
- Class ii = 5MPa (7 MPa preliminary laboratory test)
- Class iii = 1.5MPa (2 MPa preliminary laboratory test)

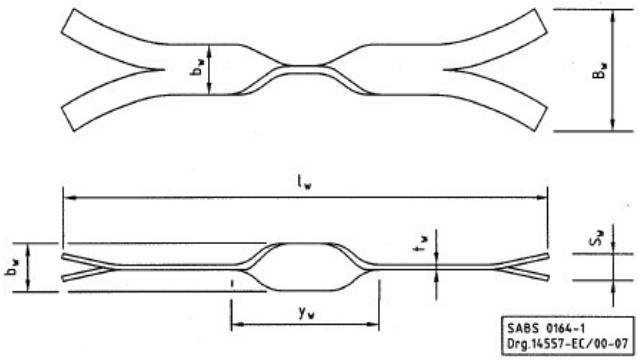
5.9 **Wall ties** shall be according to SANS 28:2010 and placed in accordance with the NHBRC, SANS 10400 and SANS 10249:2012 requirements. Note that certain plastic ties in larger cavities do not conform to the NHBRC

requirements. **A general guideline is shown below**: Note that **additional** ties shall be provided in cavity walls at vertical centres that do not exceed 300 mm centres within 150 mm of the edges of all openings and movement or control joints (or both) and where a leaf intersects with another wall. Ties shall be laid in the mortar as the work proceeds and shall have a length of embedment in the mortar of not less than 50 mm. Ties shall be set level or with a slight slope to the outside leaf of the masonry.

Spacing of wall ties

1	2	3	4	5
Nominal thickness of leaf mm	Cavity width mm	Min of ties per square meter of face	Maximum tie spacing mm	
			Vertical	Horizontal
65	50-75	5	450	450
90 or more	50-75	2,5	900	450
90 or more	75-150	3[a]	600	450

Note 3[a] - ensure the strip ties are the vertical twist type as below or any other similar type having at the same strength and stiffness



5.10 Two continuous layers of b**rickforce** are to be laid in the first two courses below a supported floor slab soffit and at the eaves wall plate level.

5.11 All brickforce to have at least **300mm overlap** at all joints.

5.12 All internal non-load bearing brickwork to stop 20mm below soffit of slabs and sealed in accordance with the Architectural, fire and acoustic Engineering specifications.

5.13 All brickwork is to be **connected** to reinforced concrete columns by means of galvanised hoop-iron straps (min 30x1.2mm thick) shot-fired into concrete through 10mm jointex at a minimum of every 4th course. Hoop-iron is to be built a minimum of 400mm into brickwork.

5.14 **Precast lintel** support (bearing) widths as per manufacturer's recommendation. As a guideline the following may be assumed initially:

- 150mm – up to 1.5m opening (span)
- 250mm – up to 2.5m opening
- 350mm – greater than 2.5m
- For lintels spanning more than 3m the Engineer is to provide specific details.
- Where lintels butt up against an RC column assume, for costing, a 100x100x10mm galvanised angle support bolted into RC column.

5.15 Lintels to have a minimum of 4 courses of **brickwork above lintel** for openings of up to a 2m span with continuous brickforce in the lower 2 courses (extending 500mm either side of opening). For openings exceeding 2m the Engineer will need to specify the minimum courses required. All lintels to remain **propped** for a minimum of 7 days after brickwork has been built.

5.16 Chasing:

- **Vertical chases** in solid units - depth not to exceed the lesser of one third of the wall/leaf thickness or 30mm. For hollow core units the depth may not exceed 15mm. Widths of chases not to exceed 70mm. Clear distance between chases should not be less than 225mm. Back to back chases need to be spaced at 225mm min as well.

- **Horizontal chases** should not exceed one sixth of the wall/leaf thickness and not exceed 1250mm in length.

- Holes and chases should **not** be made by iMPact methods in order to avoid cracking of brickwork and plaster.

- All chasing to be filled solidly with mortar once conduits are in place.

- Chasing of brickwork above openings which exceed 1m in span should be discussed with the Engineer.

5.17 Brickwork **movement joints** will be detailed by the Engineer and/or Architect at construction stage. For planning and estimation purposes allow the following:

- Clay brick walls joints at 7m spacing
- Cement brick wall at 5m spacing
- Control joints shall be at least 10 mm wide and if filled shall be filled with a compressible material such as jointex. (Soft board is not a compressible material and is unsuitable joint filler material). Typically external joints shall be sealed against the filler or a backing strip in accordance with the requirements of below.
- Joints around door and window frames, control joints, articulation joints, abutting joints at external columns and other points where **sealing** is indicated or required shall be primed, if necessary, and filled in accordance with the manufacturer's recommendations with an appropriate sealant of a colour specified in the scope of work, if any.

5.18 Sealing compounds for the building & construction industry to be in accordance with the requirements of: SANS 11077-1984 and comprise of a polyurethane-base such as Sika-pro 2hp universal one component polyurethane joint sealant or similar approved.

6.0 SURFACE BEDS

6.1 All **surface beds** and sub-surface (layerworks) preparation are to be in accordance with the details provided on the Engineering drawings. The surface finish and level tolerances are specified elsewhere.

6.2 The contractor shall provide **results of compaction** testing of the earthworks layers for approval before casting surface beds. For initial estimate purposes where no information has been supplied, assume 150mm G5 layer or 300mm of approved granular fill compacted to 98% mod aashto.

6.3 For high level **initial estimation purposes** assume a 350 micron DPC below surface beds turned up to the full depth of surface bed as bond breaker against all brickwork on all sides. Jointex to be provided around all internal columns. Assume 5% of surface beds to be reinforced with mesh 245. Assume top of surface bed level to be a minimum of 150mm above the final outside levels as per SANS 10400.

6.4 All surface bed **mix designs** must be approved by the Engineer. The mix design should be developed between the contractor and concrete supplier to ensure that all the design strength criteria are met whilst considering the curing and timing of saw-cutting.

6.5 **Saw-cut joints** will be required in all surface beds. For planning and costing purposes allow for joints to be provided at 4m centres. The aspect ratio of the panel sizes should be no more than 1:1.25. The timing of saw-cutting, which is a function of mix design and weather conditions, is left solely in the hands of the main contractor. No cracked surface bed panels will be accepted.

6.4 The concrete on which the **screed** is to be laid should have a characteristic strength of at least 20 MPa and be free of random cracking, dust, oil or other contamination. The day before laying of the screed, the base concrete should be tested for absorptiveness by pouring a cupful of water onto the surface.

If during the next few minutes it is clear that water is being absorbed the suction of the concrete should be regarded as being high. In that case the procedure is:

a) Wet the area on which the new screed or topping is to be laid and keep it wet for four hours or more.
b) Remove all free water on the surface.
c) Allow the surface to become visibly dry so that the base achieves a saturated surface-dry condition.
d) Apply the screed to the surface. On the other hand, if the test water is not visibly absorbed by the concrete during the first few minutes after application, the suction of the concrete should be regarded as being low and the screed may be applied to the dry concrete.

Concrete with 28-day characteristic strengths of 25 MPa and over, if properly compacted and cured, can be regarded as non-absorptive.

7.0 STRUCTURAL STEEL

7.1 All structural steel to **be fabricated and erected** in accordance with the relevant requirements of SANS 10162, SANS 1067, SANS 044, SANS 2001-CS1 and SANS 1921-3.

7.2 All dimensions and levels should be checked on site prior to issuing shop drawings. Any discrepancies should be brought to the attention of the Engineer and Architect. Shop drawings will only be checked for compliance with the design intent.

7.3 A complete set of shop drawings shall be submitted to the Engineer, Architect and sheeting specialist sub-contractor for approval before fabrication. Shop drawings to take sheeting requirements into account.

7.4 For measurement purposes the following finish can be assumed:

- All cold rolled steel members to be pre-galvanised
- All internal steel to receive 1 coat of primer (40micron red oxide or similar approved) and 2 coats of polyurethane enamel (40microm min per coat) to Architect's specification.
- All external steel to be hot-dip galvanised.

7.5 Where temporary bracing or propping is required the contractor shall be responsible for the design, erection, maintenance and removal of such supports.

8.0 STRUCTURAL TIMBER

8.1 Gang nailed roof trusses to be designed by Mitek or similar approved. Mitek is to inspect erection of the roof trusses and sign off that they are satisfied with the erection.

8.2 All roof trusses are to be tied down by means of galvanised hoop-iron built in securely at least 6 brick courses below wallplate as per SANS 10400.

8.3 All South African pine timber specified is to be grade 7 unless specified otherwise.

8.4 Galvanised steel strap anchors shall be taken up over the top of the rafter or tie beam, bent down on either side and nailed down from both sides, or galvanised roof ties shall be made up of two strands of wire which shall be taken up on either side of the rafter beam, twisted together so as to have no slack, but not so as to overstrain the wire, and the free ends then nailed down to prevent untwisting.

8.5 Any roof truss, rafter or beam shall be fixed to any wall by using one of the following types of anchors:

(i) Type A: two strands of 4mm galvanised steel wire;
(ii) Type B: 30mm x 1,2mm galvanised steel strap;
(iii) Type C: 30mm x 1,6mm galvanised steel strap.